

Paul Abrams

Toronto, Ontario

Senior Software Engineer · React / TypeScript · NestJS / Node.js · REST APIs & Cloud-Native
Microservices · Application & Cloud Security (OAuth 2.0 / OIDC, SAST / DAST, CWE / CVSS) · MCP &
LLM-Assisted Engineering · 15+ Years

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Professional Summary

Senior software engineer with 15+ years across the full SDLC, shipping **B2B and B2C web applications** from front-end UI through service back ends and databases. Front end in **React + TypeScript** (component-based SPAs, HTML5 / CSS3, responsive and WCAG-accessible); back end designing and implementing **RESTful APIs in TypeScript with NestJS and Node.js / Express**, plus deep **.NET Core / C#** experience on **cloud-native microservices**. **Security is a throughline: OAuth 2.0 / OpenID Connect / PKCE, JWT auth**, migration to **Azure Managed Identities, Key Vault** secret hygiene, TLS / certificate management, client- and server-side **CORS** policy, **SonarQube (SAST)** and **OWASP ZAP (DAST)** in CI pipelines, **CWE / CVSS-graded security audits**, and **NIST-based (ITSG-33 / PBMM) controls**. Hands-on with **AI-assisted engineering**: designed and shipped a **Model Context Protocol (MCP) tool-server** with in-code guardrails, develops daily with **Claude Code**, and evaluates LLM-generated code professionally. Delivers **well-tested code (Jest / Vitest / React Testing Library, xUnit)** through **PR-gated CI/CD** in **Agile / Scrum** (Professional Scrum Master I), collaborating in **Azure DevOps** with distributed stakeholders. Active open-source contributor — **60+ public PRs** including Microsoft's `vsts-team-calendar`, `tfs-cli`, and the **Microsoft Authentication Library .NET** docs.

Professional Experience

Abrams Software Toronto, Ontario (Remote)

August 2025 – Present

Full-Stack Developer · Independent

- **Chess Attack Visualization** — **React 19 + TypeScript** single-page app (Vite, **React Testing Library** + **Vitest / Jest** unit tests) consuming the **Lichess.org REST API**, with a **NestJS / Node.js** auth service implementing the **OAuth2 PKCE** flow with client- and server-side **CORS** configuration. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm.
- **CVForge** — full-stack SaaS-style product (public at github.com/pabrams/cvforge): a REST Web API (**ASP.NET Core, EF Core, Swagger / OpenAPI docs**) consumed by **two parallel SPA front ends (Angular and Vue 3)**. Designed and shipped a **Model Context Protocol (MCP) tool-server in Node.js** wrapping the API so LLM agents (**Claude Code**) can drive the domain — with **security guardrails enforced in code** (agents cannot modify locked records; agent-created content is always flagged as draft) and a **secret-scanner blocking credential leakage** into exported documents.
- **TWNR** — **architected and shipped a public-facing B2C multiplayer game, solo**. Dockerized **TypeScript** monorepo: **Node.js (Express) + WebSocket** server backed by **PostgreSQL**, per-player **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct state via database transactions. Live at twnr.paulabrams.ca.
- **LLM code evaluation** for [DataAnnotation.tech](#) and [Stellar.ai](#) — prompt fixtures and grading rubrics across **TypeScript, React, Node / NestJS, .NET / C#, Python, SQL, and MongoDB**, including **security audits with CWE / CVSS-graded vulnerability inventories**. **60+ public PRs** to outside repos including the **Microsoft Authentication Library .NET** docs and [canada-ca/platform-security](#).

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Engineer — Cloud-Native Microservices (Benefits Delivery Modernization)

- Enhanced and maintained a **cloud-native microservices** enterprise onboarding application — **.NET Core / C#** back end, **Blazor** front end, **Azure SQL, Azure Functions, GraphQL**. Designed API

endpoints, defined schemas, and generated documentation (Swagger / OpenAPI, Postman); **published and governed APIs through Azure API Management** with authentication, throttling, and transformation policies; integrated enterprise IAM via **Entra ID / Azure AD (OAuth 2.0, OpenID Connect)** with SSO / MFA.

- **Application and cloud security:** migrated the application's downstream identity from rotating secrets (Key Vault + PATs on 90-day rotation) to **Azure Managed Identities** — eliminating the secret-rotation incident class. Implemented **NIST-based ITSG-33 controls** — TLS / certificate management and **Key Vault** secret hygiene — for security assessment submissions against a **PBMM** profile; configured **Azure AD App registrations** and client- and server-side **CORS** policy.
- Designed and shipped **custom web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO REST API, and a people-by-project reporter that replaced an unusably slow query with an in-memory hashmap. Responsive UI, custom CSS, **WCAG**-aligned interactions with cross-browser support.
- Operated **CI/CD pipelines** (Azure DevOps) for build, unit testing, **Docker** containerization, **Kubernetes** deployment, and ETL orchestration; daily Git workflow with feature branches and **peer code review on PR-gated CI**. Ran weekly **demos and design walkthroughs**, mentoring junior developers on API integration and security configuration. Supported production — troubleshooting deployed services and resolving user-reported issues.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Development Lead — Internal Web Application, CI/CD Automation & DevSecOps

- Co-architected and developed an **internal C# / .NET web application** — web UI for authoring structured automation scenarios, **SQL Server** schema design, and a **co-designed REST API** providing bidirectional integration with **Azure DevOps pipelines** (versioned script-fetch and structured results-push endpoints).
- **DevSecOps / security scanning:** integrated **SonarQube** static analysis (SAST) into CI pipelines and wrote automation that **queried the SonarQube REST API to flag vulnerabilities and surface them into Azure DevOps**; ran **OWASP ZAP** dynamic scans (DAST) in pipelines.
- Engineered **Azure DevOps CI/CD pipelines** that provisioned agents to run parallel automated **Selenium** runs against deployed applications and fed structured results back into work items. Performed **code review and standards enforcement** for the team; drove a tooling consolidation that cut licensing costs.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead

- Led full **Agile / Scrum** delivery — sprint planning, review, retrospective, daily stand-up, backlog refinement, iteration demos, and design walkthroughs; introduced poker-planning estimation and capacity forecasting. **Scoped engineering requirements and timelines** with BAs, product owners, and stakeholders, and **communicated blockers and delays before they required escalation**.
- Implemented and deployed **custom web extensions in JavaScript / React / TypeScript / Node.js** with responsive, accessibility-aligned UIs — hands-on while guiding the team. **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, and execution-plan analysis.
- Administered departmental **Azure DevOps Server / TFS** instances and heavily customized process templates; **managed user/group security** and source-control access across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Development Lead — iCase / Microsoft Dynamics CRM Modernization

- Dev lead on the live **Legal Case Management System (iCase)** — a large **n-tier ASP.NET / C#** enterprise web application used by **5,000+ lawyers and contracted agents** — while concurrently building its **Microsoft Dynamics CRM** successor.

- Engineered **Dynamics CRM plug-ins, custom actions, and workflows in C#**; produced **entity-relationship diagrams** and **detailed effort-estimation breakdowns**. Established **code-review processes gated on work items** — enforcing maintainability, security, and performance standards — and mentored junior developers.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer · Release Engineering Lead

- Took over a release-management environment from a contractor team; **decommissioned over-engineered legacy desktop tooling** and replaced it with lightweight **PowerShell calling REST APIs**, bringing everything under source control.
- Implemented and deployed **web extensions in JavaScript / React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension and **contributed back upstream**. Installed, configured, and upgraded the Azure DevOps Server / TFS platform across dev / UAT / production, including **Kerberos / SPN security configuration**.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer — Legal Case Management System (iCase)

Core developer on a **distributed, multi-tier ASP.NET / C# enterprise web platform** used by **5,000+ government lawyers and contracted agent firms across Canada** — **B2B and B2C** flows (internal lawyers, external agent firms tracking their own billing) — deployed across **load-balanced web servers and regionally distributed document servers** with role- and object-based security and **24/7 high availability**.

- Led design and largely solo implementation of the **“My iCase” personalization portal** — the customizable widget-based entry point to iCase — built on **ASP.NET Web Parts with a custom SQL personalization provider, Knockout.js MVVM** (a direct ancestor of modern React component patterns), and async AJAX service calls.
- **Performance and reliability engineering**: diagnosed a homepage report taking ~10 s uncached and replaced live joins with a nightly-populated flat table (**~50x faster**); built a **self-initiated availability-monitoring suite** across the distributed servers; resolved a **COM Interop memory leak** on the document tier.
- Architected a **Java-based automation framework** that **parsed English-language scenario documents into executable scripts with round-trip code generation** — replacing a two-week, ~20-25-person manual pre-release validation cycle with automated execution reviewed by one person (**~98% reduction in effort, ~\$160K saved per release cycle**).

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 – May 2003

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

Languages: TypeScript, JavaScript, C#, SQL, HTML5 / CSS3, Python, PowerShell

Front-End: React 19, TypeScript, HTML5 / CSS3, Vite, Angular, Vue 3, Blazor, Knockout.js MVVM · **reusable, scalable component design** · responsive, cross-browser, **WCAG accessibility**

Back-End / APIs: Node.js, NestJS, Express · **.NET Core / C#, ASP.NET, Entity Framework** · **RESTful API design** — endpoints, schemas, and generated documentation (Swagger / OpenAPI, Postman) · GraphQL, WebSocket · API gateway (Azure API Management)

Architecture: cloud-native microservices, event-driven / serverless workflows, n-tier SaaS platforms, load balancing, entity-relationship modelling, design patterns

Security: OAuth 2.0 / OpenID Connect / PKCE, JWT, Entra ID / Azure AD IAM, Managed Identities, Key Vault secret hygiene, TLS / certificate management, CORS policy, SonarQube (SAST), OWASP ZAP (DAST), CWE / CVSS-graded security audits, NIST-based (ITSG-33 / PBMM) controls, secret-scanning guardrails

AI / LLM Tooling: Model Context Protocol (MCP) tool-server design with in-code agent guardrails · **Claude Code** (daily development driver) · professional **LLM code evaluation** (prompt fixtures, grading rubrics)

Databases: PostgreSQL, MS SQL Server, Azure SQL — schema design, stored procedures, indexing, execution-plan tuning · MongoDB (working familiarity)

Cloud / DevOps: Microsoft Azure (App Service, Functions, SQL, API Management, Key Vault, Managed Identities, Entra) · **AWS** (self-hosted pipeline agent + foundational training) · GCP (production hosting) ·

Docker, Kubernetes · Git, GitHub Actions, Azure DevOps Pipelines (YAML) — feature branches, PR-gated CI

Testing: Jest, Vitest, React Testing Library, xUnit / NUnit, Playwright, Selenium

Process / SDLC: Agile / Scrum (PSM I) — sprint planning, backlog refinement, demos · **peer code review**, mentoring, technical documentation · Azure DevOps, JIRA, Slack-style distributed collaboration